

Cover Letter for ERC Resubmission - P550/06/2026

KNH-UoN Ethics and Research Committee

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Date: 26 August 2026

The Secretary

KNH-UoN Ethics and Research Committee

Dear Prof. Amugune,

RE: RESUBMISSION OF REVISED RESEARCH PROPOSAL - P550/06/2026

Title: *A Hybrid Deep Learning Framework for Intracranial Aneurysms Detection: Algorithmic Development and Fairness Evaluation for Kenyan Public Health Settings*

I acknowledge receipt of the KNH-UoN ERC review letter dated 5 August 2026 (Ref: KNH-ERC/RR/561mm) and thank the committee for the thorough and constructive feedback. I have carefully addressed each of the observations and required revisions raised. The revised proposal (3 copies + Application Form) is enclosed.

Below is a summary table of how each comment has been addressed, with the page number(s) where the changes have been made. **All substantive revisions are highlighted in bold within the body of the proposal** for ease of reference.

Summary Table
of ERC Comments
and Actions
Taken

#

ERC Comment

Action Taken

Page(s) in Revised Proposal

1.1

Supervisory Team - Consider adding a radiology supervisor

Added Dr. Kevin Ombati (Department of Radiology, AKUH) as Local Data Co-Investigator with defined roles in the revised proposal (Section 1.5). The application form Personnel Block has been updated to reflect this addition.

p. 1, revised Application Form

1.2.i

Abstract - Remove references

Abstract fully rewritten without in-text citations. All citations removed.

p. iii (Abstract)

1.2.ii

Abstract - Restructure with subtitles (Background, Broad Objective, Methodology, Utility of the Study)

Abstract restructured under the four required subtitles. Each subtitle contains a concise, citation-free paragraph.

p. iii (Abstract)

2.1

Section 1.3 - Objectives must be brief, precise, and SMART

Three objectives rewritten as single-sentence SMART objectives (Specific, Measurable, Achievable, Relevant, Time-bound). Objectives 4 and 5 (interview-based) have been removed from scope.

p. 5

2.2

Section 1.4 - Research Questions must be precise

Two research questions rewritten with explicit variables, populations, outcomes, metrics, and timeframes. Research Question 3 (clinical utility/stakeholder interviews) has been removed from scope.

p. 6

Chapter 2 - Literature Review must follow the four-part flow (Burden → Past → Innovations → Why Us)

Chapter 2 fully restructured: (2.1) Public Burden of Intracranial Aneurysms, (2.2) Historical Diagnosis and Treatment and Gaps in LMICs, (2.3) Innovative Approaches: AI for Aneurysm Detection, (2.4) Why This Study? Contribution to Knowledge.

pp. 10-23

4.1

Section 3.1 - Clarify “mixed-methods” meaning (qualitative + quantitative)

Section 3.1 rewritten to explicitly state the quantitative arms (RSNA-ICA training + AKUH local validation) with experimental design. Health systems feasibility assessment reframed as technical and workflow analysis using published PACS infrastructure data and institutional knowledge of the named site lead. No interviews conducted. The qualitative/stakeholder-interview arm has been removed from the protocol.

p. 23

4.2

Section 3.2 - Name and justify study sites

Sites explicitly named and justified: (a) AKUH - primary local validation site, with formal justification for selection (diverse patient population, digital PACS archive, formal in-kind support arrangement with Dr. Kevin Ombati); (b) KENET CHUI HPC - GPU training environment; (c) UoN DoM and DCI - analysis and research environment. KNH is explicitly stated as NOT engaged as a data source (originally proposed KNH site lead has left the institution; KNH Radiology did not commit to data release within the study window). Health systems assessment addressed through technical analysis of published institutional data and the named site lead’s knowledge.

p. 23

4.3

Section 3.2 - Define study population and address consent

Study population defined for each arm: (a) RSNA-ICA training: international de-identified historical CTA scans (open-access, no consent required); (b) AKUH local validation: 200-300 retrospective adult CTA scans (Jan 2020-Dec 2025); waiver of individual consent requested from AKUH ERC. No interviews conducted.

p. 25

4.4

Section 3.3 - Inclusion and Exclusion criteria

New sub-section 3.3 added with explicit inclusion and exclusion criteria for each data source (RSNA-ICA and AKUH). Criteria cover age, imaging modality, scan quality, prior treatment, and documentation status.

pp. 25-26

4.5

Section 3.3 - Sample size justification

Sample size for AKUH local validation (n=200-300) justified using precision-based estimation: at 3-5% prevalence, expected 6-15 positive cases; sensitivity estimation precision (95% CI width +/-30% at 6 positive cases) explicitly stated. RSNA-ICA training set is the entire open-access dataset (~4,000 studies).

p. 25

4.6

Section 3.4 - Is all data retrospective?

Section 3.4 explicitly states: (a) RSNA-ICA and AKUH arms are 100% retrospective (no prospective patient contact). (b) The qualitative/stakeholder-interview arm has been entirely removed from the protocol. The health systems feasibility assessment is addressed as a technical and workflow analysis drawing on published data and the site lead's institutional knowledge.

p. 29

4.7.i

Risk of Study - AKUH data identifiability at source

New sub-section 3.7.1 added acknowledging: AKUH data is identifiable at source before de-identification; downstream use carries indirect patient-safety implications; the study is developmental only, NOT diagnostic-in-use.

p. 40

4.7.ii

Jurisdictional Gap - AKUH ERC + MTA/DTA

New sub-section 3.7.2 added: (a) AKUH ERC submission planned within one week of KNH-UoN ERC approval; (b) AKUH-UoN Material/Data Transfer Agreement (MTA/DTA) under negotiation; signed copy to be appended to final submission or provided as conditional commitment; (c) KNH-UoN ERC role is administrative (host-institution oversight) only; KNH is not a data source.

p. 41

4.7.iii

Informed Consent - Waiver justification for AKUH data

New sub-section 3.7.3 added: detailed waiver justification explaining impracticality of contacting 200-300 historical patients, HIPAA Safe Harbor 18-identifier de-identification protocol, separation of linking key held only by AKUH data custodian (Dr. Kevin Ombati), no return of individual findings, minimal risk, public benefit justification, and patient notification/opt-out mechanism.

p. 42

4.7.iv

Risk-Benefit & Non-Deployment Safeguard - Incidental findings escalation protocol

New sub-section 3.7.4 added with explicit: (1) Declaration that no AI output will inform patient care during study period. (2) Incidental Findings Escalation Protocol with 6-step table (trigger, action, responsible party, timeline). (3) Limitations of the protocol clearly stated.

p. 43

4.7.v

Stakeholder Engagement - Interviews constitute separate human subjects research

Addressed by removing the interview arm entirely from the protocol. No stakeholder interviews are conducted. The health systems feasibility assessment is addressed through technical and workflow analysis using published institutional data and the named site lead's knowledge, as described in Section 3.1. No separate consent form for interviews is required (Section 3.7.5 has been removed from the proposal).

pp. 23, 29

All changes are highlighted in **bold** in the body of the revised proposal for ease of reference.

Sincerely,

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cc: Prof. Peter Waiganjo (Primary Supervisor, DCI, UoN)

cc: Dr. Pamela Mandela (Co-Supervisor, UoN)

cc: Dr. Kevin Ombati (Local Data Co-Investigator, AKUH Radiology)

cc: Chair, Department of Mathematics, UoN
cc: Dean, Faculty of Science and Technology, UoN

Enclosures (3 hard copies each):

Revised Research Proposal (P550/06/2026)

Revised ERC Application Form

Similarity Report (updated)

Annex C: Draft AKUH-UoN Material/Data Transfer Agreement (MTA/DTA)

Annex D: AKUH ERC Submission Plan and Supporting Documentation

(Note: The interview-related annexes have been removed from the submission package as the qualitative/stakeholder-interview arm has been entirely removed from the protocol.)